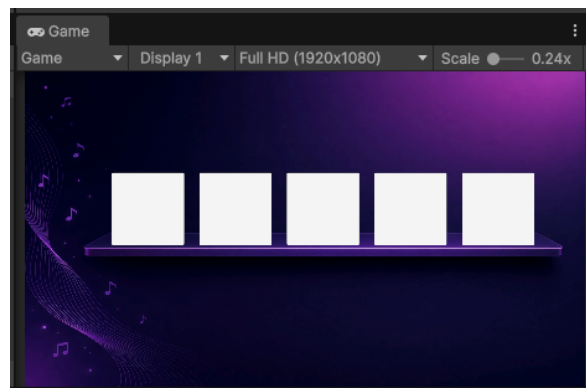
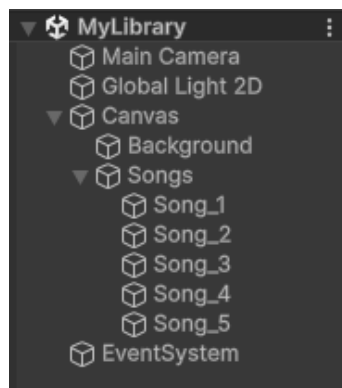


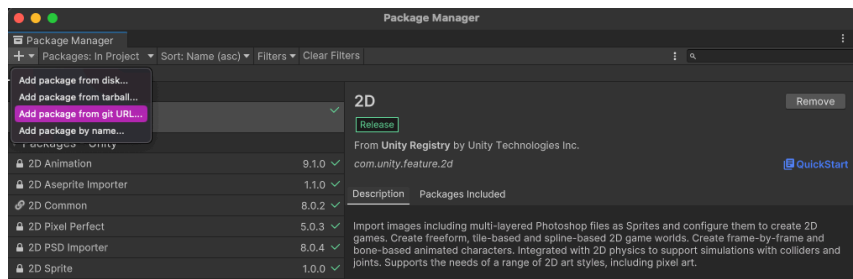
Práctica Integradora - VJ

Part #1 - Using an API to get data from a DB

1. Before you start, make sure you can get the information of your songs using your API.
2. Leave the API running.
3. Create a new project called '**PracticalIntegradora**'.
4. Rename the scene to '**MyLibrary**'.
5. Set the size of the Game window to '**FullHD**'
6. Add a **Canvas** to the scene and select the render mode as **Camera**.
7. Create a scenery of a library or a shelf to put the song covers.
8. Add one button for each one of your songs and place them on the shelf (inside an empty object). Remove the text for all of them and add an '**Horizontal Layout Group**'.



9. Go to Window -> Package Manager -> Add package from git URL:



10. Use the url: **com.unity.nuget.newtonsoft-json**

11. Create an empty script and write the **structure** that your first **store procedure** is giving you back. (The one that gives you back all the songs and their information).
12. For example '**Song**' (empty script):

```
public class Song
{
    public int idCancion {get; set;}
    public string titulo {get; set;}
    public string imagen {get; set;}
}
```

13. Create a script called "**ForceAcceptAll**" (empty script) with the following code:

```
using UnityEngine.Networking;

public class ForceAcceptAll : CertificateHandler
{
    protected override bool ValidateCertificate(byte[] certificateData)
    {
        return true;
    }
}
```

14. Create a new script called '**SelectCover**' and write the following to it:

```
using UnityEngine.UI;
using UnityEngine.Networking;
using Newtonsoft.Json;

Texture2D image;
Sprite newSprite;
public Image newImage;
```

```

public SongNo songNumber = SongNo.song1;
public enum SongNo
{
    song1, song2, song3, song4, song5
}

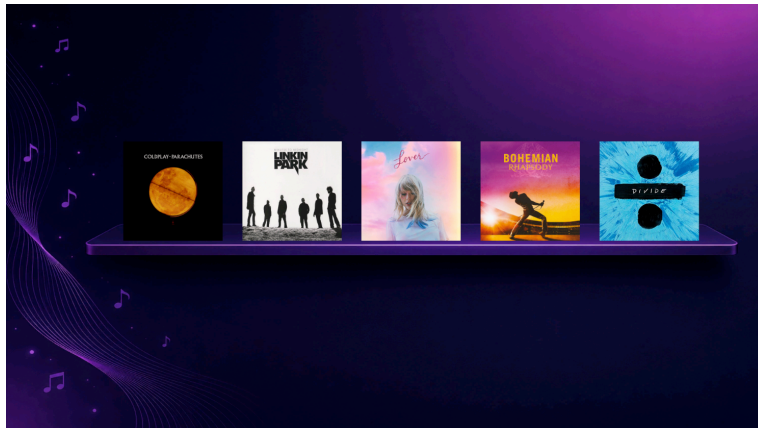
IEnumerator DownloadImage(string MediaURL)
{
    UnityWebRequest request = UnityWebRequestTexture.GetTexture(MediaURL);
    yield return request.SendWebRequest();
    if(request.result != UnityWebRequest.Result.Success)
    {
        Debug.Log(request.error);
    }
    else
    {
        image = ((DownloadHandlerTexture)request.downloadHandler).texture;
        newSprite = Sprite.Create(image, new Rect(0.0f, 0.0f,
            image.width, image.height), new Vector2(0.5f, 0.5f), 100.0f);
        newImage.sprite = newSprite;
    }
}

IEnumerator Start()
{
    string JSONurl = "https://192.168.3.14:5001/cancion";

    UnityWebRequest web = UnityWebRequest.Get(JSONurl);
    web.certificateHandler = new ForceAcceptAll();
    yield return web.SendWebRequest();
    if (web.result != UnityWebRequest.Result.Success)
    {
        UnityEngine.Debug.Log("Error API: " + web.error);
    }
    else
    {
        List<Song> songList = new List<Song>();
        songList = JsonConvert.DeserializeObject<List<Song>>
            (web.downloadHandler.text);
        string cover = songList[(int) songNumber].imagen;
        StartCoroutine(DownloadImage(cover));
    }
}

```

15. As a recommendation, **fix your store procedure** if it is not giving you back the information needed. Remember to adjust according to your structure.
16. Drag the script to each button, select its public variables and run the project.
17. **You must be able to see the covers for each song album.**



18. Create a new script called '**SongController**' and add the following to it:

```
using TMPro;
using UnityEngine.Networking;
using Newtonsoft.Json;
```

```
static public SongController Instance;
public SelectCover SelectCover;

public TextMeshProUGUI nameText;
public TextMeshProUGUI artistText;

public int SongSelection;
```

```
private void Awake()
{
    Instance = this;
    Instance.SetReferences();
    DontDestroyOnLoad(this.gameObject);
}
```

```
void SetReferences()
{
    if(SelectCover == null)
        SelectCover = FindAnyObjectByType<SelectCover>();
}

public void ChooseSong(int songIndex)
{
    SongSelection = songIndex;
    StartCoroutine(GetData());
}

IEnumerator GetData()
{
    string JSONurl = "https://192.168.3.14:5001/cancion";

    UnityWebRequest web = UnityWebRequest.Get(JSONurl);
    web.certificateHandler = new ForceAcceptAll();
    yield return web.SendWebRequest();

    if (web.result != UnityWebRequest.Result.Success)
        UnityEngine.Debug.Log("Error API: " + web.error);
    else
    {
        List<Song> songList = new List<Song>();
        songList = JsonConvert.DeserializeObject<List<Song>>
            (web.downloadHandler.text);
        LoadSongInfo(SongSelection, songList);
        PlayerPrefs.SetInt("song_no", SongSelection);
    }
}
```

```

public void LoadSongInfo(int idSong, List<Song> songList)
{
    string song = songList[idSong - 1].titulo;
    PlayerPrefs.SetString("song_name", song);
    nameText.text = song;
    string artist = songList[idSong - 1].nombre;
    PlayerPrefs.SetString("artist", artist);
    artistText.text = artist;
}

```

19. Same as before, make sure you **adapt the code** to your structure.
20. Drag the script to an empty object, select its public variables.
21. You will need to create two texts: “**Title**” and “**Artist**”.
22. Choose the function for all buttons by dragging the SongController object. **Verify functionality.**
23. Change the font for both of the texts on the screen and add a title for your scene.

